

1. What is DFS and BFS?

- **DFS (Depth-First Search):**

DFS is a graph traversal algorithm that starts from a source node and explores as far as possible along each branch before backtracking. It uses a **stack** (often via recursion) to keep track of nodes. It's useful in solving problems like maze solving, topological sorting, and cycle detection in graphs.

- **BFS (Breadth-First Search):**

BFS is a graph traversal technique that explores all the nodes at the present depth level before moving on to nodes at the next depth level. It uses a **queue** and is commonly used in shortest path problems and level order traversals.

2. What is a Recursive Algorithm?

A **recursive algorithm** is one that solves a problem by breaking it down into smaller subproblems of the same type. It continues to call itself with simpler inputs until it reaches a base condition. It is especially useful in problems like factorial, Fibonacci, tree traversals, and backtracking.

3. Explain DFS and BFS with an Example.

Example Graph:

mathematica

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A

/ \

B C

/ \

D E

- **DFS Traversal:** $A \rightarrow B \rightarrow D \rightarrow E \rightarrow C$
Explores deeply into B's subtree before visiting C.
 - **BFS Traversal:** $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$
Explores all neighbors of A first, then their neighbors.
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4. Explain A Algorithm.*

A* is an informed search algorithm used in pathfinding and graph traversal. It combines the advantages of Dijkstra's algorithm and Greedy Best-First Search. It uses a cost function:

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$$f(n) = g(n) + h(n)$$

Where:

- $g(n)$ = actual cost to reach node n
- $h(n)$ = estimated cost from n to goal (heuristic)

It is widely used in AI applications like GPS navigation, game maps, and robotics.

5. Formula of A*

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$$f(n) = g(n) + h(n)$$

- **$f(n)$** : Estimated total cost of the path through node n
 - **$g(n)$** : Cost from start to node n
 - **$h(n)$** : Heuristic cost estimate from n to the goal
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6. What is Heuristic Function?

A **heuristic function** is used in search algorithms to estimate the cost of the cheapest path from a given node to the goal. It helps algorithms like A* and Greedy Best-First Search to make decisions about which path to follow. A good heuristic reduces computation time and improves efficiency.

7. What is Greedy Search Algorithm?

A **greedy search algorithm** expands the node that appears to be closest to the goal using only the heuristic value $h(n)$. It does not consider the path cost so far ($g(n)$). It is faster but not always optimal or complete.

8. Use of Minimal Spanning Tree (MST) Algorithm

MST connects all the vertices in a weighted graph with the **minimum possible total edge weight** without forming cycles.

Applications:

- Network design (e.g., LAN, WAN)
- Clustering
- Circuit design

Popular algorithms: **Kruskal's and Prim's**

9. Explain N-Queen Problem with Example.

The **N-Queen problem** is a classic AI and backtracking problem where you must place N queens on an N×N chessboard such that no two queens attack each other (i.e., no two queens share the same row, column, or diagonal).

Example: 4-Queens Solution:

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. Q . .

. . . Q

Q . . .

. . Q .

Uses backtracking to try safe positions for each queen.

10. Explain Backtracking Process.

Backtracking is a **recursive** problem-solving method where solutions are built incrementally and abandoned ("backtracked") when they are found invalid.

Applications: N-Queen, Sudoku, Maze, Crossword solving.

11. What is a Chatbot?

A **chatbot** is an AI application that simulates human conversation through text or voice. Chatbots use **NLP (Natural Language Processing)** to understand and respond to user inputs.

Examples: ChatGPT, Siri, Alexa, Google Assistant

12. Explain DES Algorithm.

DES (Data Encryption Standard) is a symmetric-key block cipher that:

- Uses a **56-bit key**
- Works on **64-bit blocks**
- Performs **16 rounds** of complex operations
- Uses substitution and permutation steps

It is now considered outdated due to its vulnerability to brute-force attacks.

13. Explain AES Algorithm.

AES (Advanced Encryption Standard) is a symmetric-key encryption standard:

- Uses key sizes of **128, 192, or 256 bits**

- Works on **128-bit blocks**
- Uses **10, 12, or 14 rounds** based on key size
- Involves operations like SubBytes, ShiftRows, MixColumns, and AddRoundKey

It is widely used in modern security systems.

14. Difference between AES and DES

Feature	DES	AES
Key Size	56-bit	128/192/256-bit
Block Size	64-bit	128-bit
Security	Less secure	More secure
Rounds	16	10/12/14
Speed	Slower	Faster
Status	Obsolete	Industry Standard

15. Explain RSA Algorithm.

RSA is an **asymmetric** cryptographic algorithm that uses:

- **Public key** for encryption
 - **Private key** for decryption It is based on the mathematical problem of **factoring large prime numbers**.
Used in secure data transmission, digital signatures.
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16. Explain Diffie-Hellman Algorithm.

A method to securely exchange **cryptographic keys** over a public channel.

- Each party generates a public and private key.
- Shared secret is computed using both public and private keys.

No key is transmitted, hence it's secure from eavesdropping.

17. Explain MD5 Algorithm.

MD5 (Message Digest 5) is a hashing algorithm that:

- Produces a **128-bit hash** value
- Used for verifying **data integrity**

- Not secure against collisions, but still used in basic checksums
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18. What is Public Key?

A cryptographic key used for **encrypting** data in asymmetric encryption. It is shared openly and used with a corresponding **private key**.

19. What is Private Key?

A cryptographic key used for **decrypting** data encrypted with the public key. It is kept **secret** by the owner.

20. What is Asymmetric Key Cryptography?

Cryptography that uses **two keys** – a public key to encrypt and a private key to decrypt.

Example algorithms: RSA, ECC

21. What is Symmetric Key Cryptography?

Encryption technique where the **same key** is used for both encryption and decryption.

Examples: AES, DES

22. Difference: Asymmetric vs Symmetric

Feature	Symmetric	Asymmetric
Keys Used	One (same)	Two (public/private)
Speed	Faster	Slower
Security	Moderate	Higher
Examples	AES, DES	RSA, ECC

23. What is Secret Key?

A **private key** used in symmetric encryption, shared between sender and receiver.

24. Asymmetric Key Algorithms:

- RSA
- ECC (Elliptic Curve Cryptography)
- DSA

- Diffie-Hellman
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25. Symmetric Key Algorithms:

- AES
 - DES
 - Triple DES
 - Blowfish
 - RC4
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26. What is Cryptography and Cryptanalysis?

- **Cryptography:** Science of securing information using encryption.
 - **Cryptanalysis:** The study of breaking cryptographic algorithms or decoding information without the key.
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27. What is Artificial Intelligence?

AI is the simulation of **human intelligence** by machines, enabling them to **think, learn, reason**, and make decisions. Fields include machine learning, NLP, robotics.

28. Real-World Applications of AI:

- Voice Assistants (Siri, Alexa)
 - Self-driving Cars
 - Healthcare Diagnosis
 - Fraud Detection
 - Chatbots and Virtual Agents
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29. What is CIA Triad?

A foundational model in information security:

- **Confidentiality:** Protecting data from unauthorized access
- **Integrity:** Ensuring data is accurate and untampered
- **Availability:** Ensuring systems are accessible when needed